

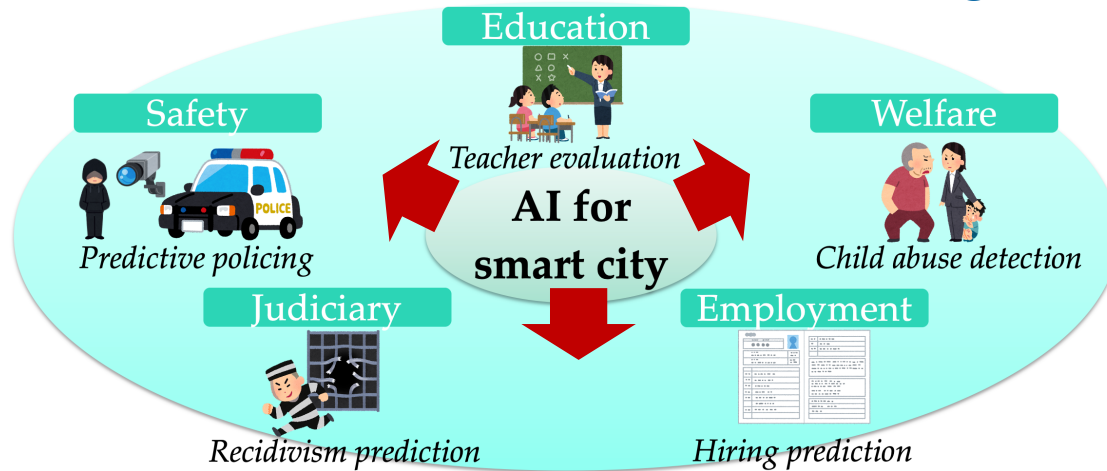


Toward Interventionally Fair Decision-Making under Partial Causal Graph Knowledge

2026/08/08

NTT Communication Science Laboratories
Yoichi Chikahara

Algorithmic Decisions via ML Prediction



Input Training data about individuals

A Gender (Sensitive)	Q Skill	D Department	M Physical strength	Y Decision
Female	A	Science	B	Accept
Male	B	Economics	B	Accept
⋮				

Output Machine learning (ML) model

$$\hat{Y} = h_{\theta}(A, Q, D, M)$$

Prediction ML model
with parameter θ

Toward Society Shaped by Algorithmic Decision-Making

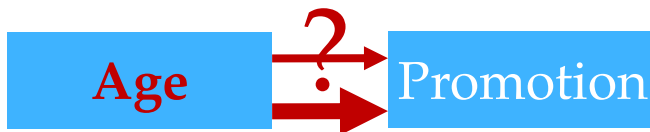
- ML predictions must be **fair** with respect to *sensitive attributes* (e.g., race, gender)
- Discrimination is often legally assessed based on causality:

An Age Discrimination Case [U.S. Supreme Court (2009)]



Plaintiff must show that **age was the *but-for cause*** of the employment decision

Are there any *causal effects*?



1. Achieve fairness using a cluster causal graph [Chikahara. **UAI2026**]

**Fairness under Graph Uncertainty: Achieving Interventional Fairness
with Partially Known Causal Graphs over Clusters of Variables**

Yoichi Chikahara¹

2. Latent discrimination discovery via mean and variance causal graph discovery [Chikahara. **KDD2026**] *(if time permits)*

**Moment Matters: Mean and Variance Causal Graph Discovery
from Heteroscedastic Observational Data**

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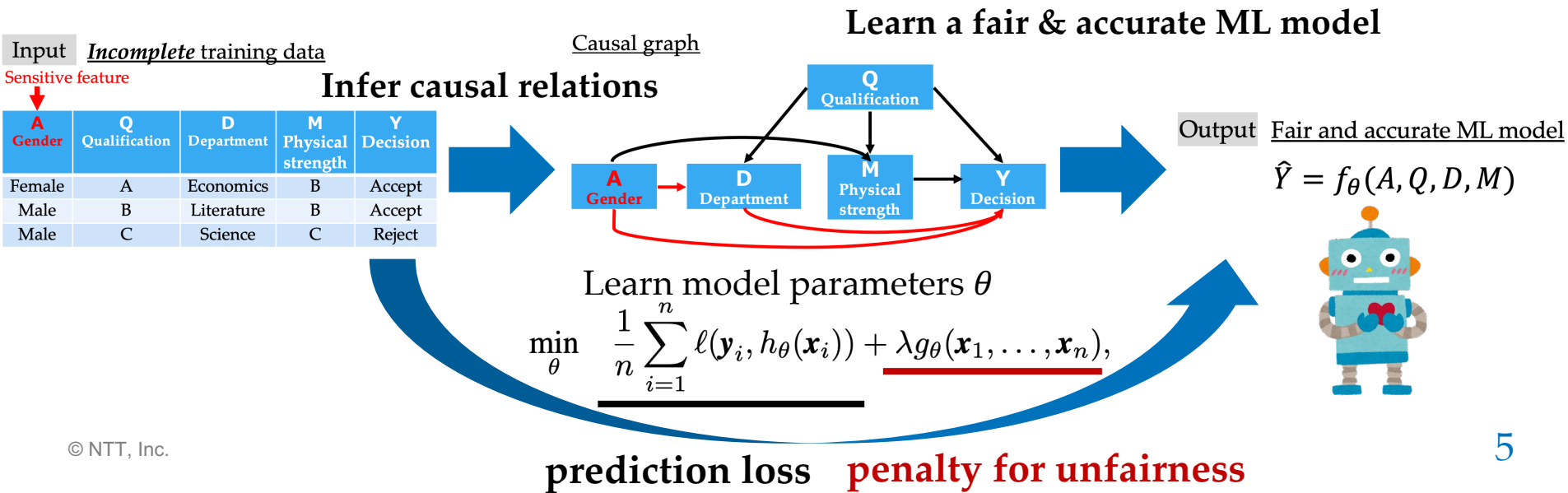
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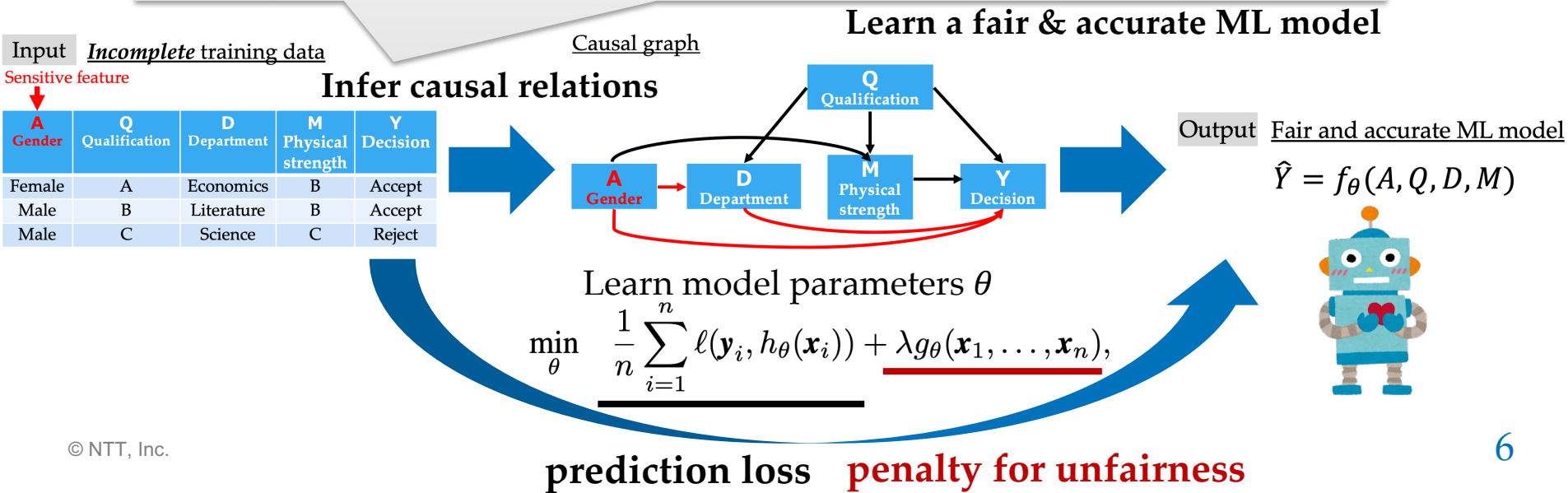
Fair & Accurate ML Prediction via Causal Inference



Fair & Accurate ML Prediction via Causal Inference

Accurately inferring a causal graph is challenging ☹️

- We are NOT interested in the causal graph itself!
- We just want to ensure fairness



Q: Can we ensure fairness using *high-level* causal graph representation?

Input

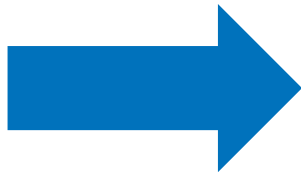
Observational data

Conditional Independence
(CI) tests over
clusters of variables

User-specified cluster partition

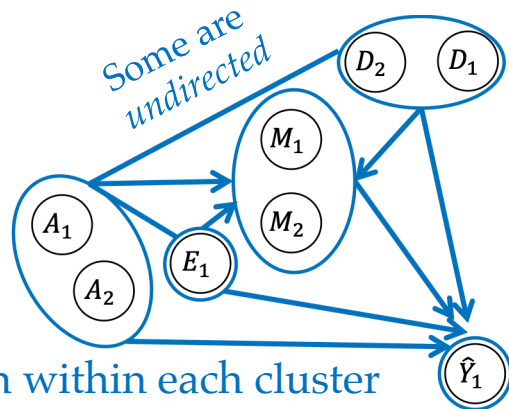
$$\mathbf{A} = [A_1, A_2], \mathbf{E} = [E_1]$$

$$\mathbf{M} = [M_1, M_2], \dots, \hat{\mathbf{Y}} = [\hat{Y}_1]$$



Cluster CPDAG

[Anand+; NeurIPS2025]



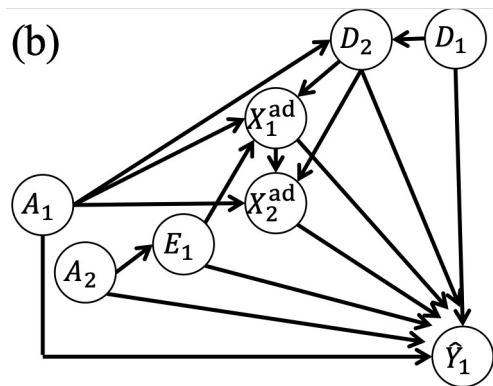
Graph within each cluster
is **unknown**

Cluster CPDAG inference is **much easier** than variable-level one 😊

- Graph size is reduced, so the number of required CI tests is much smaller

Example scenario [Chikahara+; AISTATS2021, DMKD2023]: Hiring decisions for physically demanding jobs

Causal graph



A: gender & race (*sensitive features*)

X^{ad} : physical test score (*admissible features*)

D: medical test results

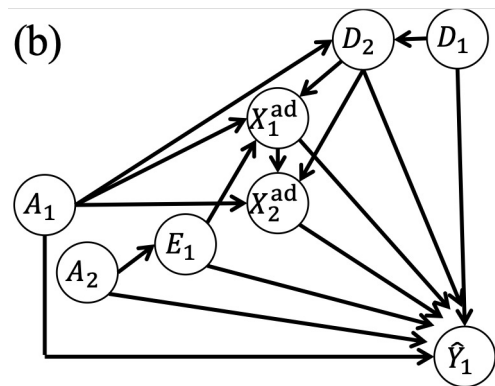
E: educational background

$\hat{Y}$: prediction

Indirect effects are **not** discriminatory

Hiring decisions for physically demanding jobs

Causal graph



A: gender & race (*sensitive features*)

X^{ad}: physical test score (*admissible features*)

D: medical test results

E: educational background

$\hat{Y}$: prediction

A structural causal model (SCM)
describes data-generating processes

E.g.,

$$X_1^{\text{ad}} = \underbrace{f_{X_1^{\text{ad}}}}_{\text{deterministic (unknown)}}(A_1, D_2, E_1, \underbrace{U_{X_1^{\text{ad}}}}_{\text{exogenous noise}})$$

deterministic
(unknown)

exogenous noise

$$\hat{Y} = \underbrace{h_{\theta}}_{\text{predictive model (known)}}(\underbrace{X, U_{\hat{Y}}}_{\text{randomness in probabilistic model}})$$

predictive model
(known)

randomness
in probabilistic model

1. Interventions: Replacing structural equations

E.g.,
$$X_1^{\text{ad}} = \underbrace{f_{X_1^{\text{ad}}}(A_1, D_2, E_1)}_{\text{deterministic (unknown)}} \underbrace{U_{X_1^{\text{ad}}}}_{\text{exogenous noise}}$$

Intervention
 $do(X_1^{\text{ad}} = x_1^{\text{ad}})$



$$X_1^{\text{ad}} = x_1^{\text{ad}}$$

Values are forced to be constant x_1^{ad}

2. Interventional fairness:

Definition 2.1 (Salimi et al. [2019]). Prediction $\hat{Y}$ is *interventionally fair* with respect to sensitive features A , if

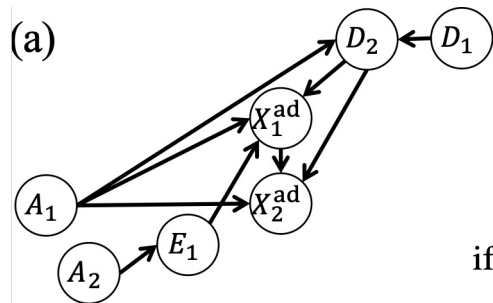
$$\begin{aligned} &P(\hat{Y} = \hat{y} \mid do(A = \mathbf{a}), do(\mathbf{X}^{\text{ad}} = \mathbf{x}^{\text{ad}})) \\ &= P(\hat{Y} = \hat{y} \mid do(A = \mathbf{a}'), do(\mathbf{X}^{\text{ad}} = \mathbf{x}^{\text{ad}})), \end{aligned} \quad (4)$$

holds for all possible prediction values $\hat{y}$, for any discrete values $\mathbf{a}, \mathbf{a}'$ of sensitive features A and for any discrete values $\mathbf{x}^{\text{ad}}$ of admissible features $\mathbf{X}^{\text{ad}}$.

Changing sensitive attributes
do not change the distributions of predictions

- Group variables according to a user-defined partition

Variable-level DAG



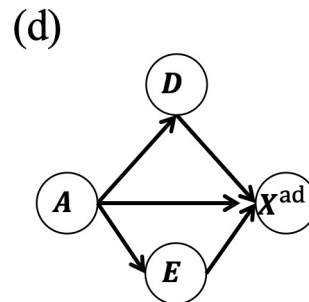
Grouping



$$\mathbf{C}_i \rightarrow \mathbf{C}_j$$

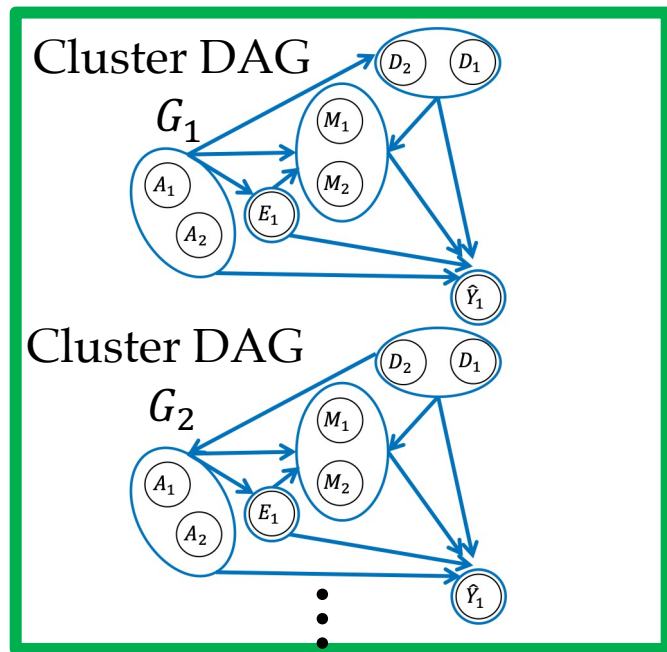
if some $V_i \in \mathbf{C}_i$ and $V_j \in \mathbf{C}_j$
have edge $V_i \rightarrow V_j$ in G_v .

Cluster DAG



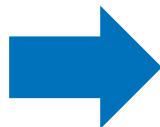
Graph structure
within cluster is
unknown

Cluster MEC

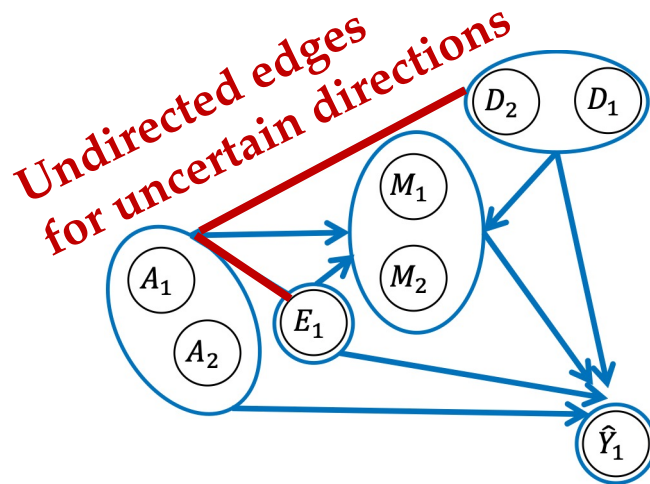


Cluster DAGs with **identical**
conditional independence relations over clusters

Single Graph
Representation



Cluster CPDAG

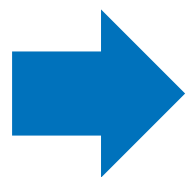


Inference by CLOC algorithm [Anand+; NeurIPS2025]

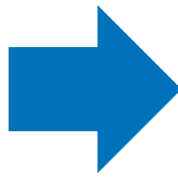
Input



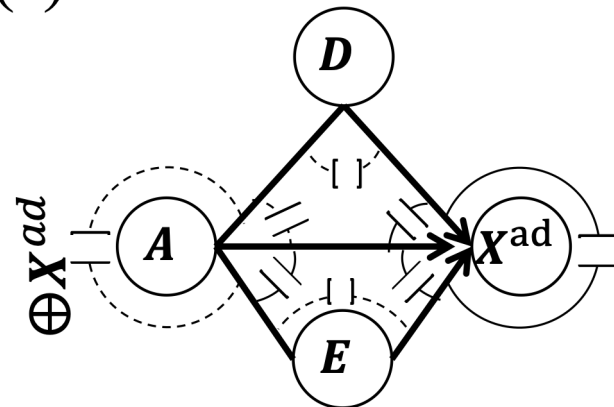
CI tests



$$\begin{aligned} A &\not\perp\!\!\!\perp D \\ A &\perp\!\!\!\perp D | X^{ad} \\ &\vdots \end{aligned}$$



(e)



User-specified cluster partition

$$\begin{aligned} A &= [A_1, A_2], E = [E_1] \\ D &= [D_1, D_2], \dots, \hat{Y} = [\hat{Y}_1] \end{aligned}$$

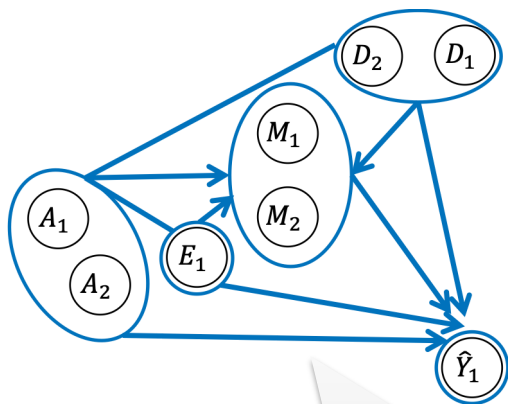
Note: Cluster-level edges are **insufficient** to represent CI relations

Arcs — --- and marks $\oplus$ $\oslash$
are needed to explicitly represent CI relations.

Q: Can we estimate the distance between interventional distributions using cluster CPDAGs?

Cluster CPDAG

[Anand+; NeurIPS2025]



Graph structure
within cluster is
unknown

Distance between interventional distributions

$$P(\hat{Y} = \hat{y} \mid do(A = \mathbf{a}), do(X^{\text{ad}} = \mathbf{x}^{\text{ad}}))$$

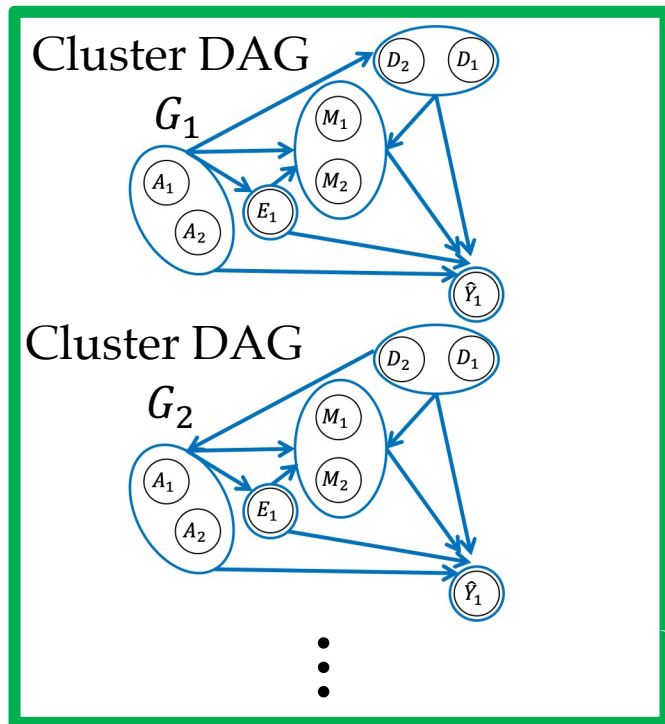


Unfairness of predictions $\hat{Y}$

$$P(\hat{Y} = \hat{y} \mid do(A = \mathbf{a}'), do(X^{\text{ad}} = \mathbf{x}^{\text{ad}}))$$

A: We can estimate the *maximum* distance over possible cluster DAG structures in the **cluster MEC**!

Cluster MEC



Distributional distances

Distance estimated with G_1

Distance estimated with G_2

Maximum

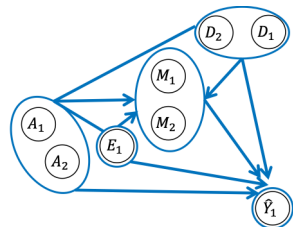
Worst-case unfairness under

graph uncertainty

Q: Can we obtain the maximum without performing DAG enumeration?

Our idea: Back-door adjustment set enumeration

Adjustment candidate sets



Cluster MEC

Z^1

Z^2

Distributional distances

Distance estimated
with Z^1

Distance estimated
with Z^2

⋮

Maximum

Worst-case
unfairness

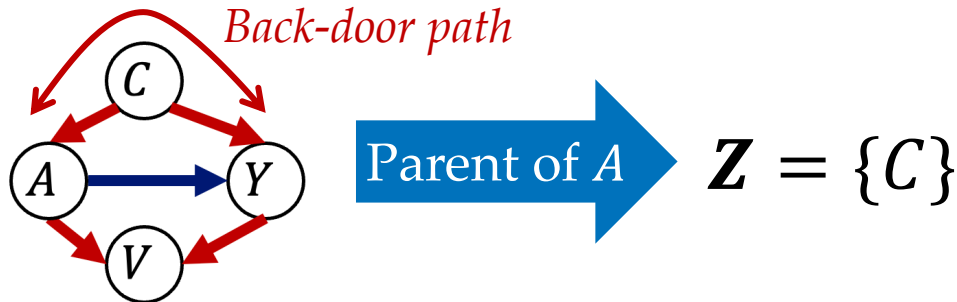
Back-door adjustment

- Adjustment set **Z**: Identifying an interventional distribution via marginalization

$$\begin{aligned} & P(\hat{Y} = \hat{y} \mid do(A = \mathbf{a}), do(X^{\text{ad}} = \mathbf{x}^{\text{ad}})) \\ &= \mathbb{E}_{\mathbf{Z}, X^{\text{re}} | \mathbf{a}, \mathbf{x}^{\text{ad}}} \left[P(\hat{Y} = \hat{y} | A = \mathbf{a}, X^{\text{ad}} = \mathbf{x}^{\text{ad}}, \mathbf{Z}, X^{\text{re}}) \right] \\ & \quad \times X^{\text{re}} = X \setminus \{A, X^{\text{ad}}, \mathbf{Z}\} : \text{remaining variables} \end{aligned}$$

- Parent set is a common choice of adjustment set:

Example:



Proposed penalty function

Adjustment candidate sets

Parent enumeration +
Cluster addition for
cluster CPDAGs

Cluster MEC

$\mathbf{Z}^1$

$\mathbf{Z}^2$

$\vdots$

MMD estimation via IPW

Kernel MMDs

MMD estimated
with $\mathbf{Z}^1$

MMD estimated
with $\mathbf{Z}^2$

$\vdots$

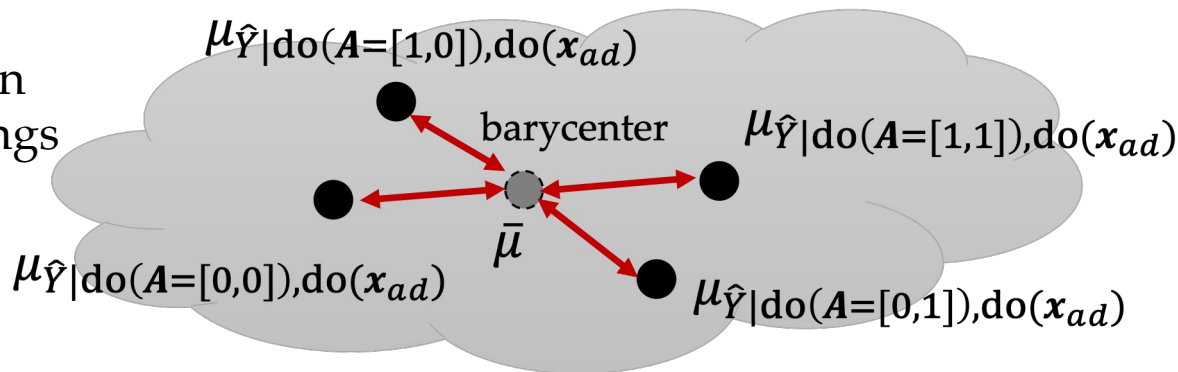
Maximum

Worst-case
unfairness

Proposed penalty function

Kernel MMD:

Distance between
kernel embeddings



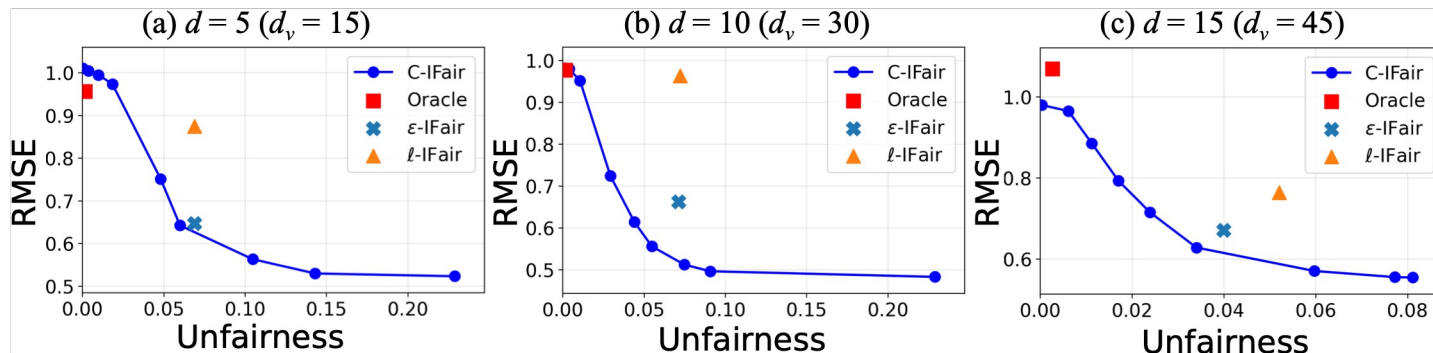
$$\begin{aligned}
 & g_{\theta}(\mathbf{x}_1, \dots, \mathbf{x}_n) \\
 &= \max_{\mathbf{x}^{\text{ad}}} \sum_{\mathbf{a}} \sum \left\| \hat{\mu}_{\hat{\mathbf{Y}}|do(\mathbf{a}),do(\mathbf{x}^{\text{ad}})} - \underbrace{\hat{\bar{\mu}}_{\hat{\mathbf{Y}}|do(\mathbf{x}^{\text{ad}})}}_{\text{barycenter}} \right\|_2^2, \\
 & \hat{\mu}_{\hat{\mathbf{Y}}|do(\mathbf{a}),do(\mathbf{x}^{\text{ad}})} = \frac{1}{n} \sum_{i=1}^n w_i \phi_{\text{RFF}}(\hat{\mathbf{y}}_i) \quad \text{where } w_i = \frac{\mathbf{I}(A_i = \mathbf{a}, \mathbf{X}_i^{\text{ad}} = \mathbf{x}^{\text{ad}})}{\pi_{\mathbf{a}, \mathbf{x}^{\text{ad}}}(\mathbf{z}_i^m)}, \text{ (IPW weight)}
 \end{aligned}$$

Weight depends on each set $\mathbf{Z}^m$

Empirical Results

Empirically, the proposed method performs best 😊

	ADULT		GERMAN		OULAD	
	AUC↑	UNFAIRNESS↓	AUC↑	UNFAIRNESS↓	AUC↑	UNFAIRNESS↓
ORACLE	0.709 ± 0.009	0.000 ± 0.000	0.582 ± 0.004	0.000 ± 0.000	0.628 ± 0.015	0.000 ± 0.000
FULL	0.874 ± 0.004	0.030 ± 0.024	0.762 ± 0.060	0.173 ± 0.011	0.697 ± 0.024	0.061 ± 0.001
UNAWARE	0.805 ± 0.049	0.018 ± 0.014	0.695 ± 0.061	0.101 ± 0.013	0.654 ± 0.019	0.004 ± 0.000
NO-DESCS	0.815 ± 0.007	0.026 ± 0.022	0.726 ± 0.059	0.085 ± 0.008	0.654 ± 0.020	0.004 ± 0.000
ϵ -IFAIR	0.812 ± 0.006	0.015 ± 0.012	0.756 ± 0.008	0.082 ± 0.000	0.641 ± 0.005	0.005 ± 0.002
ℓ -IFAIR	0.764 ± 0.002	0.015 ± 0.008	0.750 ± 0.035	0.151 ± 0.011	0.639 ± 0.011	0.003 ± 0.001
C-IFAIR	0.829 ± 0.021	0.014 ± 0.010	0.760 ± 0.061	0.065 ± 0.008	0.660 ± 0.018	0.001 ± 0.000



<https://arxiv.org/abs/2602.23611>

This work was supported by JST ACT-X

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Q: Can causal graph discovery help discover *latent* sensitive attributes?



latent sensitive attributes

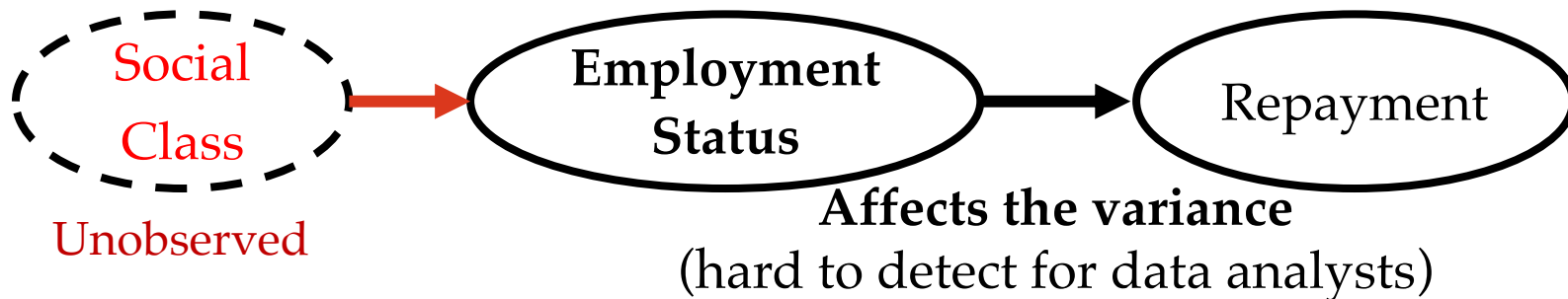
Legally acceptable for now,
but potentially discriminatory
in the future!

X_1	X_2	X_3	...	Y
				Accept
				Reject
				Reject
				Reject

A large number of feature attributes

- Features that affect the (conditional) variance may cause *latent* statistical discrimination
 - ✓ Avoiding subgroups with high outcome variance is common for risk aversion, but decision rules for such purpose might be discriminatory

Example:

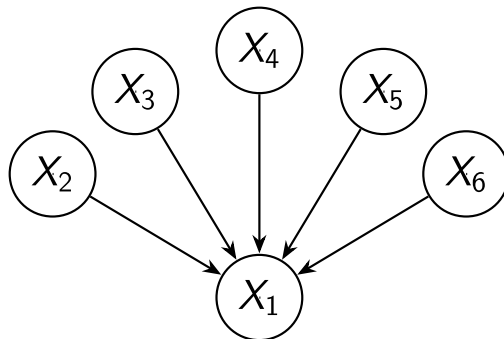


Heteroscedasticity: $\text{Var}[R|E]$ is not a constant but a function of E

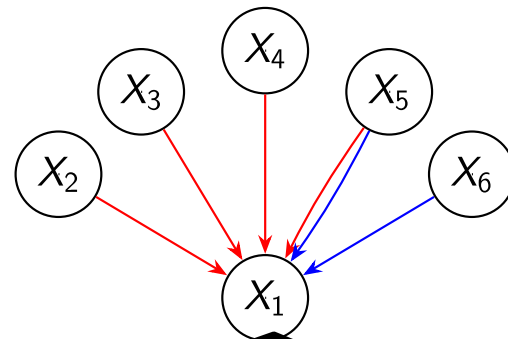
Moment-driven causal discovery



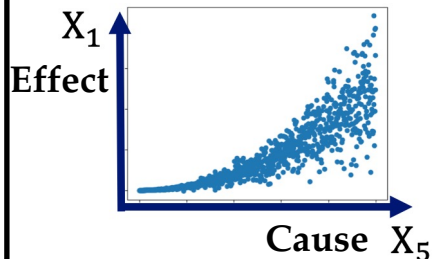
Standard causal graph



Mean/**Variance** Causal Graphs



Heteroscedastic data:



Defined by extending identifiable SCM class called heteroscedastic noise models (HNMs)

- **Mean-Variance HNMs** [Chikahara; KDD2026]:

$$X = f_X \left(\text{pa}^M(X) \right) + g_X(\text{pa}^V(X)) U_X$$

Bayesian causal discovery based on identifiability

Input Observational data D

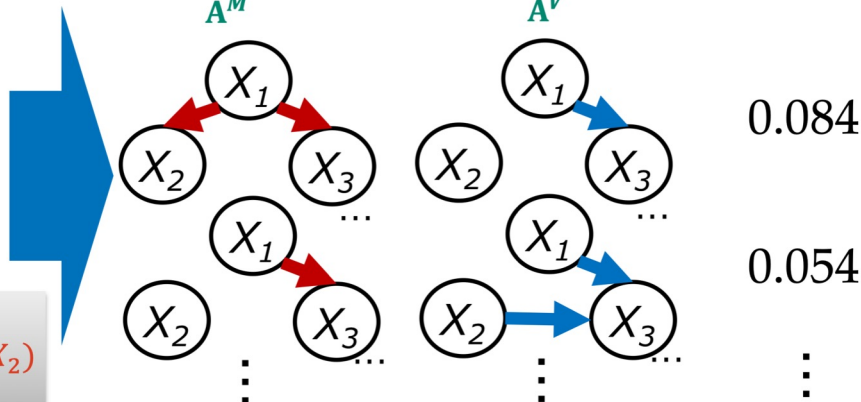
X_1	...	X_d

Input Prior knowledge about true causal graph structure

Edge of G^M	Edge of G^V	Order
$X_1 \rightarrow X_3$	$X_1 \rightarrow X_3$	$\pi(X_1) < \pi(X_2)$
$\vdots$	$\vdots$	$\vdots$

Output Posterior over causal graphs $P(\mathbf{A}^M, \mathbf{A}^V | D)$

Mean causal graph Variance causal graph Probability



- Advantage:** Can evaluate probability on structural feature of causal graphs

$$\mathbb{E}_{P(\mathbf{A}^M, \mathbf{A}^V | D)}[f(\mathbf{A}^M, \mathbf{A}^V)]; \text{ e.g., } f(\mathbf{A}^M, \mathbf{A}^V) = \mathbb{I}(X_i \rightarrow X_j \text{ in } \mathbf{A}^M) \text{ or } \mathbb{I}(X_i \rightsquigarrow X_j \text{ in } \mathbf{A}^V) \text{ (path)}$$

Theorem 3.5: Under Assumptions 3.1, 3.2, 3.3, and 3.4, mean and variance causal graphs G^M and G^V are identifiable from observational distribution $P(\mathbf{X})$ if for $j = 1, \dots, d$, (A) m_j is a nonlinear function, (B) v_j is a piecewise function, **but not a constant function**, and (C) E_j is a Gaussian noise.

Take-home messages

1. Cluster causal graphs enable fair and accurate decision-making under graph uncertainty.
2. Mean–variance causal discovery can help identify hidden sources of discrimination through interpretable causal graph representations.

Thank you!

